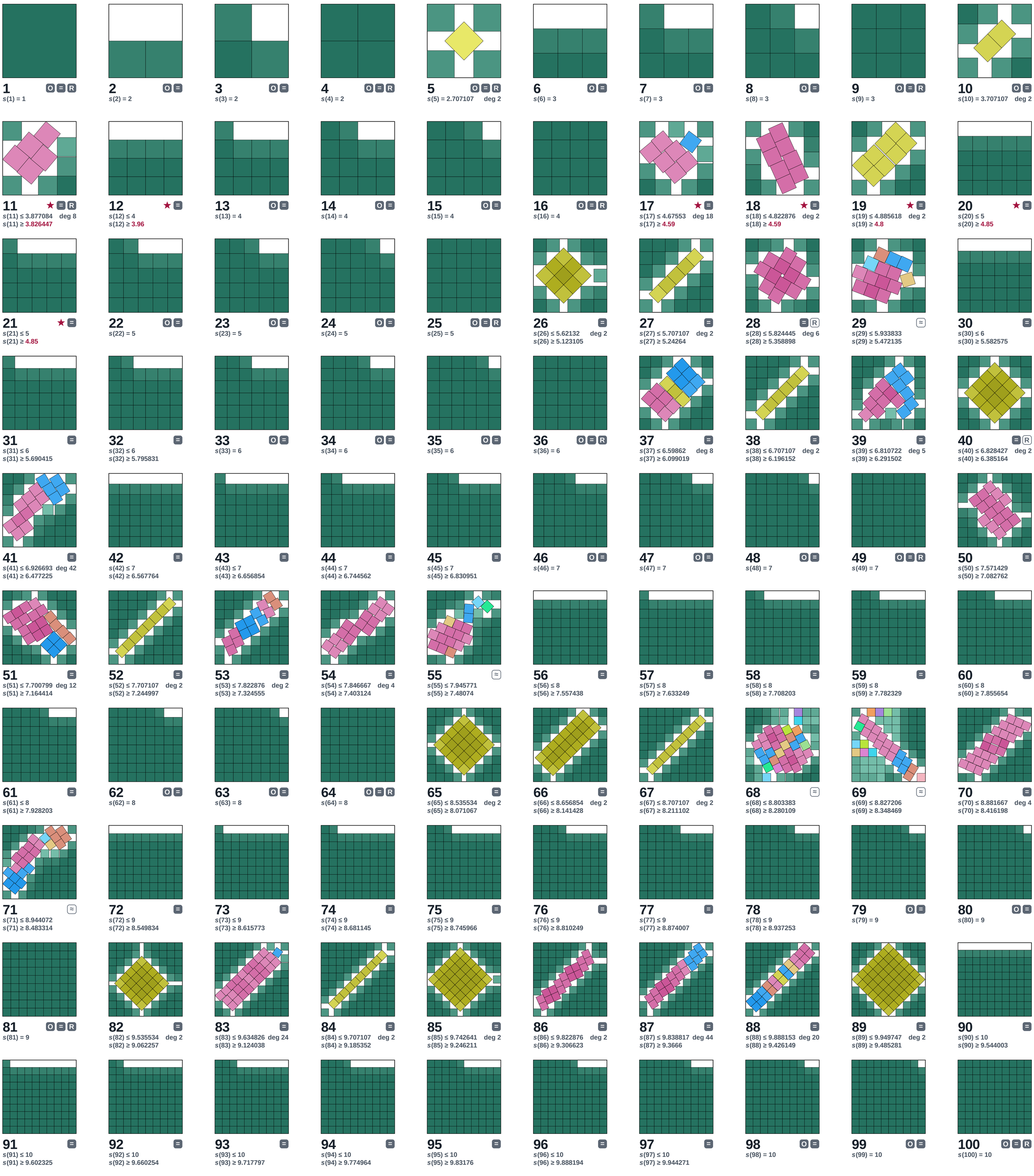
100 BEST KNOWN SQUARE PACKINGS

★ **Including new results (September 8, 2026)**

github.com/jlevy/squares



1 O = R
s(1) = 1

2 O =
s(2) = 2

3 O =
s(3) = 2

4 O = R
s(4) = 2

5 O = R
s(5) = 2.707107 deg 2

6 O =
s(6) = 3

7 O =
s(7) = 3

8 O =
s(8) = 3

9 O = R
s(9) = 3

10 O =
s(10) = 3.707107 deg 2

11 ★ = R
s(11) ≤ 3.877084 deg 8
s(11) ≥ 3.826447

12 ★ =
s(12) ≤ 4
s(12) ≥ 3.96

13 O =
s(13) = 4

14 O =
s(14) = 4

15 O =
s(15) = 4

16 O = R
s(16) = 4

17 ★ =
s(17) ≤ 4.67553 deg 18
s(17) ≥ 4.59

18 ★ =
s(18) ≤ 4.822876 deg 2
s(18) ≥ 4.59

19 ★ =
s(19) ≤ 4.885618 deg 2
s(19) ≥ 4.8

20 ★ =
s(20) ≤ 5
s(20) ≥ 4.85

21 ★ =
s(21) ≤ 5
s(21) ≥ 4.85

22 O =
s(22) = 5

23 O =
s(23) = 5

24 O =
s(24) = 5

25 O = R
s(25) = 5

26 =
s(26) ≤ 5.62132 deg 2
s(26) ≥ 5.123105

27 =
s(27) ≤ 5.707107 deg 2
s(27) ≥ 5.24264

28 = R
s(28) ≤ 5.824445 deg 6
s(28) ≥ 5.358898

29 ≈
s(29) ≤ 5.933833
s(29) ≥ 5.472135

30 =
s(30) ≤ 6
s(30) ≥ 5.582575

31 =
s(31) ≤ 6
s(31) ≥ 5.690415

32 =
s(32) ≤ 6
s(32) ≥ 5.795831

33 O =
s(33) = 6

34 O =
s(34) = 6

35 O =
s(35) = 6

36 O = R
s(36) = 6

37 =
s(37) ≤ 6.59862 deg 8
s(37) ≥ 6.099019

38 =
s(38) ≤ 6.707107 deg 2
s(38) ≥ 6.196152

39 =
s(39) ≤ 6.810722 deg 5
s(39) ≥ 6.291502

40 = R
s(40) ≤ 6.828427 deg 2
s(40) ≥ 6.385164

41 =
s(41) ≤ 6.926693 deg 42
s(41) ≥ 6.477225

42 =
s(42) ≤ 7
s(42) ≥ 6.567764

43 =
s(43) ≤ 7
s(43) ≥ 6.656854

44 =
s(44) ≤ 7
s(44) ≥ 6.744562

45 =
s(45) ≤ 7
s(45) ≥ 6.830951

46 O =
s(46) = 7

47 O =
s(47) = 7

48 O =
s(48) = 7

49 O = R
s(49) = 7

50 =
s(50) ≤ 7.571429
s(50) ≥ 7.082762

51 =
s(51) ≤ 7.700799 deg 12
s(51) ≥ 7.164414

52 =
s(52) ≤ 7.707107 deg 2
s(52) ≥ 7.244997

53 =
s(53) ≤ 7.822876 deg 2
s(53) ≥ 7.324555

54 =
s(54) ≤ 7.846667 deg 4
s(54) ≥ 7.403124

55 ≈
s(55) ≤ 7.945771
s(55) ≥ 7.48074

56 =
s(56) ≤ 8
s(56) ≥ 7.557438

57 =
s(57) ≤ 8
s(57) ≥ 7.633249

58 =
s(58) ≤ 8
s(58) ≥ 7.708203

59 =
s(59) ≤ 8
s(59) ≥ 7.782329

60 =
s(60) ≤ 8
s(60) ≥ 7.855654

61 =
s(61) ≤ 8
s(61) ≥ 7.928203

62 O =
s(62) = 8

63 O =
s(63) = 8

64 O = R
s(64) = 8

65 =
s(65) ≤ 8.535534 deg 2
s(65) ≥ 8.071067

66 =
s(66) ≤ 8.656854 deg 2
s(66) ≥ 8.141428

67 =
s(67) ≤ 8.707107 deg 2
s(67) ≥ 8.211102

68 ≈
s(68) ≤ 8.803383
s(68) ≥ 8.280109

69 ≈
s(69) ≤ 8.827206
s(69) ≥ 8.348469

70 =
s(70) ≤ 8.881667 deg 4
s(70) ≥ 8.416198

71 ≈
s(71) ≤ 8.944072
s(71) ≥ 8.483314

72 =
s(72) ≤ 9
s(72) ≥ 8.549834

73 =
s(73) ≤ 9
s(73) ≥ 8.615773

74 =
s(74) ≤ 9
s(74) ≥ 8.681145

75 =
s(75) ≤ 9
s(75) ≥ 8.745966

76 =
s(76) ≤ 9
s(76) ≥ 8.810249

77 =
s(77) ≤ 9
s(77) ≥ 8.874007

78 =
s(78) ≤ 9
s(78) ≥ 8.937253

79 O =
s(79) = 9

80 O =
s(80) = 9

81 O = R
s(81) = 9

82 =
s(82) ≤ 9.535534 deg 2
s(82) ≥ 9.062257

83 =
s(83) ≤ 9.634826 deg 24
s(83) ≥ 9.124038

84 =
s(84) ≤ 9.707107 deg 2
s(84) ≥ 9.185352

85 =
s(85) ≤ 9.742641 deg 2
s(85) ≥ 9.246211

86 =
s(86) ≤ 9.822876 deg 2
s(86) ≥ 9.306623

87 =
s(87) ≤ 9.838817 deg 44
s(87) ≥ 9.3666

88 =
s(88) ≤ 9.888153 deg 20
s(88) ≥ 9.426149

89 =
s(89) ≤ 9.949747 deg 2
s(89) ≥ 9.485281

90 =
s(90) ≤ 10
s(90) ≥ 9.544003

91 =
s(91) ≤ 10
s(91) ≥ 9.602325

92 =
s(92) ≤ 10
s(92) ≥ 9.660254

93 =
s(93) ≤ 10
s(93) ≥ 9.717797

94 =
s(94) ≤ 10
s(94) ≥ 9.774964

95 =
s(95) ≤ 10
s(95) ≥ 9.83176

96 =
s(96) ≤ 10
s(96) ≥ 9.888194

97 =
s(97) ≤ 10
s(97) ≥ 9.944271

98 O =
s(98) = 10

99 O =
s(99) = 10

100 O = R
s(100) = 10

O proved optimal (35) = exact value known (95) ≈ only known numerically (5) R rigid (established here) (12) R annotated rigid by the catalogue (2) ★ lower bound first proved here (7)

colors indicate distinct tilt angles 4 3 2 1 0 shade indicates number of full-side contacts

s(*n*) is the side of the smallest square holding *n* unit squares; deg is the algebraic degree of that side length

Diagram by Joshua Levy with assistance from Claude and Codex

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